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Studierichting: Informatica

Oefeningenreeks 2D nr. 11.14

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{\sqrt{3x-1} - \sqrt{5x-7}}{\sqrt{13-2x} - \sqrt{3x-2}} \\ &= \lim_{x \rightarrow 3} \frac{((3x-1) - (5x-7)) (\sqrt{13-2x} + \sqrt{3x-2})}{(\sqrt{3x-1} + \sqrt{5x-7}) ((13-2x) - (3x-2))} \\ &= \lim_{x \rightarrow 3} \frac{(-2x+6) (\sqrt{13-2x} + \sqrt{3x-2})}{(\sqrt{3x-1} + \sqrt{5x-7}) (15-5x)} \\ &= \lim_{x \rightarrow 3} \frac{-2(x-3) (\sqrt{13-2x} + \sqrt{3x-2})}{-5(x-3) (\sqrt{3x-1} + \sqrt{5x-7})} \\ &= \lim_{x \rightarrow 3} \frac{2 (\sqrt{13-2x} + \sqrt{3x-2})}{5 (\sqrt{3x-1} + \sqrt{5x-7})} \\ &= \frac{2 (\sqrt{7} + \sqrt{7})}{5 (\sqrt{8} + \sqrt{8})} \\ &= \frac{4\sqrt{7}}{10\sqrt{8}} \\ &= \frac{\sqrt{14}}{10} \end{aligned}$$

References